Danyang Jiang

PhD Student in Astrophysics

Kavli Institute for Astronomy and Astrophysics & School of Physics, Peking University

jiangdy@stu.pku.edu.cn | (+86) 157 6267 6930

ORCID: 0009-0003-6747-2221 | Homepage: https://dyjiang.github.io/

Education

Peking University, Beijing, China — Sep 2024 - Jul 2029 (expected)

Ph.D. in Astrophysics, Kavli Institute for Astronomy and Astrophysics (KIAA)

Advisor: Prof. Linhua Jiang

Peking University, Beijing, China — Sep 2020 - Jul 2024

B.S. in Astronomy; Advisor: Prof. Linhua Jiang

Undergraduate thesis: Cosmic Evolution of the Fe II/Mg II Line Ratio in Quasar Broad-Line Regions

Research Highlights

Research interests:

- High-redshift Quasar, AGN and LRD
- Early black hole growth and galaxy-AGN coevolution
- Large-scale environment of AGNs
- The Epoch of Reionization
- JWST emission-line spectroscopy and wide-field surveys

Publications (as of November 1, 2026)

- A total of **14** papers in major astronomical journals (ApJ, Nature Astronomy, etc.), including **6** first-authored papers (2 in preparation).
- A total of **95** citations based on the SAO/NASA ADS.

Research Skills

Observations	JWST/NIRCam imaging and WFSS; JWST/NIRSpec and VLT/XSHOOTER spectroscopy; HST, Subaru/HSC, SDSS, and DESI data analysis
Methods	Spectral analysis and emission-line measurements; AGN-host decomposition; Environment and clustering analyses; SED fitting; injection-recovery completeness simulations; Slitless spectroscopic data reduction and forward modeling; maximum-likelihood estimation and Bayesian inference (MCMC)

Teaching

- Teaching Assistant, Foundations of Theoretical Physics I, Peking University, 2025
- Teaching Assistant, Atomic Physics, Peking University, 2026

Honors and Scholarships

- CAST Young Science and Technology Talent Program – PhD Special Program, China Association for Science and Technology, 2025-2027
- President Scholarship, Peking University, 2024 & 2025 & 2026

Selected Conferences

- **Academic Annual Conference of Chinese Astronomical Society** 2026
  Shijiazhuang, China
  Talk: *Large-scale Environmental Dependence of AGN Activity Revealed by JWST Paα Emitters*
- **EREBUS / COSMOS-3D Meeting** 2026
  Ann Arbor, MI, USA
  Talk: *Paschen-line AGNs: MBH, luminosity functions, and large-scale environments*
- **The 28th Guo Shoujing Symposium on Frontiers in Galaxy Cosmology** 2026
  Haikou, China
  Talk: *Black Hole Mass Estimators and Luminosity Functions of Paschen-line AGNs*
- **EREBUS / COSMOS-3D Meeting** 2025
  Bologna, Italy
  Talk: *Ruling out AGNs as the dominant source of cosmic reionization with JWST*
- **The 27th Guo Shoujing Symposium on Frontiers in Galaxy Cosmology** 2025
  Changsha, China
  Talk: *Ruling out AGNs as the dominant source of cosmic reionization with JWST*
- **European Astronomical Society Annual Meeting** 2025
  Cork, Ireland (online)
  Poster: *No Redshift Evolution in the Fe II/Mg II Flux Ratios of Quasars across Cosmic Time*

Publications and Manuscripts

First-authored papers

[1] **Jiang, D.**, Jiang, L., Fu, S., et al. (2026), "COSMOS-3D: Black Hole Mass Estimators and Luminosity Functions of Paschen-line AGNs", *The Astrophysical Journal*, 1004, 92.

[2] **Jiang, D.**, Jiang, L., Fu, S., et al. (2026), "Large-scale Environmental Dependence of AGN Activity Revealed by JWST Paα Emitters", submitted to *The Astrophysical Journal Letters,* arxiv.org/abs/2610.19320.

[3] **Jiang, D.**, Jiang, L., Sun, S., et al. (2025), "AGNs ruled out as the dominant source of cosmic reionization", *Nature Astronomy*, 9, 1890-1897.

[4] **Jiang, D.**, Onoue, M., Jiang, L., et al. (2024), "No Redshift Evolution in the Fe II/Mg II Flux Ratios of Quasars across Cosmic Time", *The Astrophysical Journal*, 975, 214.

[5] **Jiang, D.**, et al. (2026), "No Evidence for Large-scale Environmental Dependence of Little Red and Blue Dots", manuscript in preparation.

[6] **Jiang, D.**, et al. (2026), "Little Red and Blue Dots: Nuclear and Host Emission Revealed by JWST Multiband Imaging", manuscript in preparation.

Co-authored papers

[1] Fu, S., Zhang, Z., **Jiang, D.**, et al. (2026), "The discovery of two little red dots in transition into quasars", *Nature Astronomy*, 10, 1372-1386.

[2] Fu, S., Zhang, Z., **Jiang, D.**, et al. (2026), "A Comprehensive JWST Slitless Spectroscopic Census of ~12,000 Emission-line Galaxies at $0 < z < 9$ in COSMOS", submitted to *The Astrophysical Journal Supplement Series*.

[3] Chen, J., Jiang, L., ..., **Jiang, D.**, et al. (2026), "Hot Molecular Hydrogen in Galaxies at Redshift ~ 0.2-1.5 Revealed by JWST", submitted to *The Astrophysical Journal*, arxiv.org/abs/2610.19319.

[4] Fu, S., Jiang, L., ..., **Jiang, D.**, et al. (2026), "A JWST View of Cosmic Star Formation from z~0 to 9 with Hydrogen Recombination Lines in the COSMOS and GOODS-N fields", submitted to *The Astrophysical Journal*, arxiv.org/abs/2610.19318.

[5] Meyer, R. A.; Wang, F., ..., **Jiang, D.**, et al. (2026), "JWST COSMOS-3D: Spectroscopic census and luminosity function of [O III] emitters at $6.75 < z < 9.05$ in COSMOS", *Astronomy & Astrophysics,* 712, 16.

[6] Li, Z.-J., Zou, S., ..., **Jiang, D.**, et al. (2026), "COSMOS-3D: Dense Circumnuclear Gas Across Black Hole Growth Phases at $z \sim 3$", *The Astrophysical Journal*, 1002, 9.

[7] Sun, S., Bosman, S. E. I., ..., **Jiang, D.**, et al. (2026), "Broad Paschen lines as black hole mass tracers: A reverberation mapping calibration", Submitted to *Astronomy & Astrophysics*, arxiv.org/abs/2608.24645.

[8] Wang, Y., Zhang, Z., ..., **Jiang, D.**, et al. (2026), "A JWST Study of Paα and Hα Emitter Mergers at $z \sim 1$-6.5 and Their Impact", manuscript in preparation.